

$$P(A|B) = \frac{P(AB)}{P(B)}, \quad P(B) > 0$$

$$AB = A \cap B$$

$$A+B = A \cup B$$

Family of two children

B: at least one boy is in the family

A: both children are boys



$$P(B) = 3/4$$

$$P(AB) = 1/4$$

$$\rightarrow P(A|B) = 1/3$$

BE: blue eyes

B: blond

$$P(BE|B) > P(BE)$$

$$\frac{P(BE \cdot B)}{P(B)} > P(BE)$$

$$P(BE \cdot B) > P(B)P(BE)$$

$$\frac{P(BE \cdot B)}{P(BE)} > P(B)$$

$$P(B|BE) > P(B)$$

multiplication rule

$$P(AB) = P(A) \cdot P(B|A)$$

A: the 1st is an ace

B: the 2nd is an ace

C: the 3rd is an ace

